ADITYA SAXENA

Pune, Maharashtra | Website | Gmail | Github | Linkedin

WORK EXPERIENCE

Aditya Birla Groups - Hindalco Industries Ltd. Renukoot , UP
Software Development Intern – AI & Analytics *May 2025 – Jul 2025*

- Engineered a production-ready AI & analytics platform for Hindalco's aluminium extrusion plant by using messy data from the PLC data pipelines, designing an explainable anomaly detection engine over **36,000+** Aluminium Extrusion manufacturing records under **Six Sigma** methods , and delivering a full-stack workflow system featuring **time-series prediction** using multiple algorithms using **FastAPI** and **SQLite** for operational decision support.

Research Experience

University of Illinois Springfield, USA & KIIT Remote
Research Collaborator *Jul 2025 – Jun 2026*

- First author of an under-review study evaluating LLM alignment under token-budget constraints through the proposed Alignment Tax metric.
- Authored an under-review capability-based authorization framework for securing LLM agents with formally defined permission enforcement.

University of West Florida, USA & KIIT Remote
Research Collaborator *Jul 2026*

- First-author of a study on prompt injection after context compaction in LLM agents, analyzing 1,894 trials across multiple open-weight models.
- Investigated capability degradation in long-horizon LLM agents under repeated context compaction, proposing Compactation Tax , a framework to quantify memory-induced performance erosion.

UNIVERSITY PROJECTS

Machine Learning Phishing Detection System [Agentic AI Pipeline, BERT, Data Analysis, FastAPI]

- Led the end-to-end ML pipeline for a routed specialist-model phishing detection system, including DistilBERT fine-tuning, automated data pipelines, ablation studies, W&B experiment tracking, and security evaluation, achieving validation F1 scores of **0.9707** (email) and **0.9248** (URL) with perfect modality routing while benchmarking classical baselines. A Graded Major Project :- Outstanding , Position :- Team Leader

RepLift – Offline-First Nutrition Tracker *Next.js, TypeScript, FastAPI, PostgreSQL, Redis, Dexie*

- Built an offline-first nutrition tracker with resilient IndexedDB synchronization (optimistic updates, idempotency keys, exponential backoff, conflict resolution) and an AI-assisted natural-language food logger featuring structured-output validation, privacy controls, and manual fallback, validated through **54 passing frontend unit tests**.

ACHIEVEMENTS & CERTIFICATIONS

Aditya Birla Hindalco Group Industries Ltd.
Hindalco Role Model – " You Inspire Me " Recognition *Jul 2025*

- Awarded under the **ABG Bhoomika – ALERT Behavior Initiative** by the Plant Head, Extrusion Downstream, for outstanding internship contributions.

KIIT University
Top 10 Finalist – Smart India Hackathon Internal Hackathon *Mar 2024*

- Led front-end development for a disaster management solution, ranking among the **Top 10** teams out of **100+** participating teams.

Additional Certifications: LinkedIn Certification Portfolio

TECHNICAL SKILLS

- **Languages & Backend:** Python, SQL, TypeScript, FastAPI, REST APIs, Git, Linux, HTML, CSS
- **AI & Machine Learning:** NLP, Transformers (BERT/DistilBERT), Agentic AI, Time-Series Forecasting, EDA
- **Frameworks & Data:** Next.js, PostgreSQL, Redis, SQLite, Pandas, NumPy, scikit-learn, Hugging Face, Weights & Biases (W&B)

EDUCATION

Kalinga Institute of Industrial Technology Bhubaneshware , Odisha
Bachelor of Engineering *Expected 2026*